Bayesian Adaptive Portfolio Optimisation Exploiting Fully Probabilistic Design

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Abstract

Web of Science reports more than eighteen thousand items concerning portfolio optimisation, which indicates the topic's importance and provokes the question of whether it makes sense to add a new item to this pile. The paper positively answers it by a single complete design. Bayesian learning of an autoregressive model with exogenous variables optimally chooses the conjugate prior probability density function of the unknown parameters, estimates the model structure by a genetic search over an exact Bayesian criterion, tracks the parameter changes via an optimised data-based forgetting, which robustifies the estimation against outlying observations, and updates the sufficient statistic safely. The learnt model then serves an expected multi-step quadratic-loss design of the decision policy, which relies on a numerically safe, feasible square-root optimisation with a receding horizon, elicits the weights of the quadratic loss by exploiting the fully probabilistic design, which handles the risks in a non-traditional but consistent way, and opens a way to an off-line design with Thompson's sampling of the parameters.

The solution is complete in the sense that every step, from the choice of the prior to the constrained design, is derived within a single framework; it can also be applied to many other decision-making problems. A paired comparison on 200 random portfolios and four market windows against equal-weight, minimum-variance and mean-variance allocations and the S&P 500, an ablation of every component and stress tests on synthetic data with known parameters are reported with the released code. The policy has consistently smaller drawdowns than mean-variance allocation and its adaptive forgetting rejects outlying observations as designed; it does not outperform the equal-weight portfolio after transaction costs, and the experiments locate the reason in the elicitation of the risk weight and in the strength of the optimal prior.

Keywords: portfolio optimisation, probabilistic modelling, adaptive decision-making, fully probabilistic design[1]

Highlights

- complete, consistent, and analytical solution of the hard trading problem
- structure estimation and hierarchical priors
- novel adaptive forgetting, rejecting outlying observations
- novel elicitation of the risk-aware quadratic loss
- numerically safe implementation and few tuning knobs
- paired benchmark, ablation and stress-test study with released code

1. Introduction

The paper presents a structured, methodologically consistent approach to portfolio optimisation. It complements, improves and combines all the steps needed for an adaptive, feasible, and complete solution. It utilizes many known practices and results from Bayesian statistics, [1, 2, 3], tracking of parameter changes, [4], linear-quadratic adaptive regulation (LQR), [5], especially in the fully probabilistic design (FPD) version, [6], quadratic optimisation [7], and an established search heuristics, [8]. The main contribution of this article is unifying, improving upon, and connecting these

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parts. The design is useful not only on financial markets but, with slight modifications, in a wide variety of dynamic decision-making problems.

On Related Work

The research in quantitative portfolio optimisation spans a vast area of different views on the problem and its solution, as recent samples from this journal illustrate [9, 10, 11].

Classical approaches [12] frame the portfolio choice as a dual optimisation, where expected returns are maximised and variance is minimised. Many innovative approaches exist, for instance [13], which tunes the objective function directly to the investor's preferences like an aversion to risk or a diversification priority. However, the complex choice among many possible configurations is left to the user's responsibility, which hinders the approach strength and makes an objective evaluation of the gain in performance hard.

The systematic way to the specific version of the common preference elicitation problems [14] is a significant part of our solution. It optimises expected quadratic loss expressing portfolio-choice objectives. Together with the adopted, online estimated, autoregressive returns model with exogenous variables (ARX), [1], the solution becomes a version of the linear quadratic regulator task, which is so common in the control domain, [15]. The use of novel adaptive forgetting [16] makes the resulting allocation more robust against black swan events, notoriously connected with financial markets [17].

In the portfolio optimisation context, the LQR setting was applied in [18] to optimise the efficient frontier, there enhanced by particle swarm optimisation. It also applies to related tasks, for example to continuous-time hedging of option portfolios, which, unlike a multitude of academic papers, considers transaction costs [19]. Other typical instances of portfolio optimisation, including machine learning, are reviewed by [20]. The range of applied methods and the variability in the results indicate that the portfolio optimisation problem is far from being solved.

Paper's Aim

This work aims to develop a solution that: • does not leave out any of its important subtasks; • is as deductive as possible; • relies as little as possible on the user's tuning of the hyperparameters.

Notation

We consider a finite number of assets $N \in \mathbb{N}$ = natural numbers. The returns of individual assets are observed in discrete time $t \in \mathbb{N} \cup \{0\}$. They are denoted as $y_t \in [-1, +\infty)^N$ with y_t being an N-vector containing the returns of all N assets indexed by $i \in \{1, \dots N\}$ $(y_{i,t})_{i=1}^N$. All of the considered assets are denominated in United States Dollar (USD) and currency risks are not considered.

We use $z_t \in \mathbb{R}^\rho$, $\rho \geq N$, $\mathbb{R}$ denotes the real line, a ρ-vector of other observed variables, possibly exogenous factors, at time t, as the regressors. $d_{i,t} \in [0, 1)$, $i \in \{1, \dots, N\}$, is the transaction cost associated with the asset i at time t.

The opted allocation of funds forms an N-vector a_t, which expresses percentages of the portfolio the respective assets occupy. This is the action at time t in our decision-making task. We consider a spot portfolio with no leverage and short-selling. This poses constraints on the action space $\{a \in \mathbb{R}^N \mid \mathbf{1}^\top a = 1, a \geq 0\}$.

The collection of all data up to the time step t is $\mathcal{D}^{(t)} = (\mathcal{D}^{(t-1)}, (y_t, z_t, d_t))$, $\mathcal{D}^{(0)}$ marks prior knowledge.

Layout

The market returns are modelled by a multivariate ARX model with unknown parameters, Sec. 2, learnt by the Bayesian methodology, [1, 25]: the choice of the prior probability density function (PDF) is in Sec. 2.1, the model-structure estimation in Sec. 2.2, and the parameter tracking with forgetting in Sec. 2.3. The newest point estimate inserted into the ARX model provides the predictor used by the policy optimisation, which is formulated as an LQR task in Sec. 3 and cares, like the estimation, about an efficient, numerically safe, square root implementation. Its loss function quantifies the user's desire to maximise expected returns while including transaction costs and respecting the uncertainty about future returns; the quantitative specification exploits the fully probabilistic design, [21], which extends classical decision-making theory, [22, 23], and stochastic control, [24], and integrates the uncertainty-induced risks, Sec. 3.1. Sec. 4 reports a paired comparison with standard portfolio benchmarks, an ablation of every component of the pipeline, a sensitivity study and stress tests; it identifies the parts that carry the performance and the two design elements that limit it. The paper is closed by comments in Sec. 5.

Pipeline

The proposed portfolio optimisation uses (mostly well-known) techniques from Bayesian statistics and

optimisation. The pipeline of our process is shown in Fig. 1.

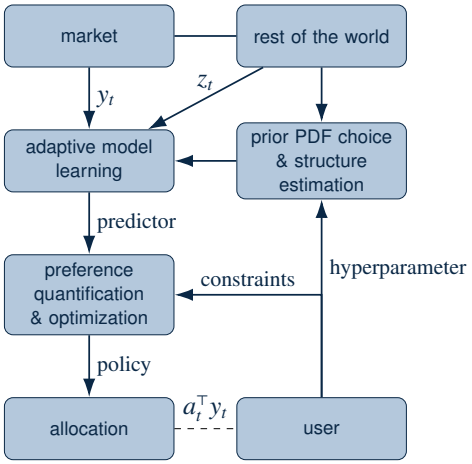


Figure 1: *Solution pipeline*: the market produces the observed returns y_t of the assets of the user's portfolio, and the rest of the world gives the information $z_t = (a_t, x_{t-1})$ potentially useful for predicting them. The data serve the structure estimation, the choice of the Bayesian prior and the forgetting. The learnt model (available knowledge) $\rightarrow y_t$ also serves the quantitative expression of the user's wishes and the optimal allocation a_t of the funds before y_t realises; afterwards the user receives the return $a_t^\top y_t$.

2. Complete Learning of the Returns Model

A multivariate autoregressive model with exogenous variables serves well for predicting market returns y_t: it is simple, it allows an efficient Bayesian parameter estimation and it gives the returns' predictor. The ARX model reads

$$y_t = \mathbf{P}^\top z_t + e_t,\ z_t^\top = [a_t^\top, x_{t-1}^\top] \quad e_t \sim \mathcal{N}(0, \mathbf{\Sigma}), \tag{1}$$

where $\mathbf{P}$ is (ρ, N)-matrix of regression coefficients, x_{t-1} is a vector of regressors. The N-vector e_t is normally distributed noise with zero mean and covariance $\mathbf{\Sigma}$. The noise e_t is independent of z_t and e_τ for $\tau < t$.

$\mathbf{P}$ and $\mathbf{\Sigma}^{-1}$ form an unknown parameter $\theta = (\mathbf{P}, \mathbf{\Sigma}^{-1}) \in \Theta$ of the model. The corresponding parametric model, PDF marked throughout $p(\cdot)$ for the join, prior, posterior and predictive PDFs, $m(\cdot)$ for the returns model, and $r(\cdot)$ for rules of the decision policy, reads $m(y_t|a_t, \mathcal{D}^{(t-1)}, \theta)$

$$= \frac{\exp\left[-\frac{1}{2}(y_t - \mathbf{P}^\top z_t)^\top \mathbf{\Sigma}^{-1}(y_t - \mathbf{P}^\top z_t)\right]}{(2\pi)^{\frac{N}{2}} |\mathbf{\Sigma}|^{\frac{1}{2}}}. \tag{2}$$

To get the predictor $p(y_{t+1}|a_{t+1}, \mathcal{D}^{(t)})$, where $\mathcal{D}^{(t)}$ marks knowledge available at time t, we use Bayes' rule

$$p(\theta|\mathcal{D}^{(t)}) = \frac{p(\mathcal{D}^{(t)}|\theta, \mathcal{D}^{(0)}) p(\theta|\mathcal{D}^{(0)})}{\int_\Theta p(\mathcal{D}^{(t)}|\theta, \mathcal{D}^{(0)}) p(\theta|\mathcal{D}^{(0)}) d\theta}. \tag{3}$$

As usual, the prior knowledge $\mathcal{D}^{(0)}$ in the prior PDF $p(\theta) = p(\theta|\mathcal{D}^0)$ and other PDFs are implicitly present.

The chain rule for PDFs provides the joint PDF of all available data conditioned on the unknown θ (and $\mathcal{D}^{(0)}$)

$$p(\mathcal{D}^{(t)}|\theta) = \prod_{\tau=1}^{t} m(y_\tau|a_\tau, \mathcal{D}^{(\tau-1)}, \theta) r(a_\tau|\mathcal{D}^{(\tau-1)}, \theta) \tag{4}$$

The parameter $\theta \in \Theta$ is unknown to the user who chooses the actions. Thus, PDFs describing the user's policy formed by randomised rules r meet the natural conditions of control, [1], $(r(a_t|\mathcal{D}^{(t-1)}, \theta) = r(a_t|\mathcal{D}^{(t-1)}))_{t=1}^{H}$, where the considered horizon $H \in \mathbb{N}$. Under them, (3) is

$$\begin{aligned} p(\theta|\mathcal{D}^{(t)}) &= \frac{l(\mathcal{D}^{(t)}, \theta) p(\theta)}{J(\mathcal{D}^{(t)})}, \text{ where the denominator is} \\ J(\mathcal{D}^{(t)}) &= \int_\Theta l(\mathcal{D}^{(t)}, \theta) p(\theta) d\theta, \text{ and} \\ l(\mathcal{D}^{(t)}, \theta) &= \prod_{\tau=1}^{t} m(y_\tau|a_\tau, \mathcal{D}^{(\tau-1)}, \theta) \end{aligned} \tag{5}$$

is the likelihood function, i.e., the factor of the PDF (4) modelling environment (at fixed $\mathcal{D}^{(t)}$) depending on the possible parameter values.

Using the likelihood function and the denominator (5) plus marginalisation, [1], we obtain the exact predictor of future returns

$$p(y_{t+1}|a_{t+1}, \mathcal{D}^{(t)}) = \frac{J(\mathcal{D}^{(t+1)})}{J(\mathcal{D}^{(t)})}. \tag{6}$$

The likelihood (5) for the PDF (2) can be expressed as

$$\begin{aligned} l(\mathcal{D}^{(t)}, \theta) &= (2\pi)^{-N\frac{t}{2}} |\mathbf{\Sigma}|^{-\frac{t}{2}} \\ \times\ & \exp\left\{ -0.5\mathrm{tr}\left[\mathbf{\Sigma}^{-1} \begin{pmatrix} -\mathbf{I} & \mathbf{P}^\top \end{pmatrix} \tilde{\mathbf{V}}_t \begin{pmatrix} -\mathbf{I} \\ \mathbf{P} \end{pmatrix}\right]\right\}. \end{aligned}$$

The extended information matrix $\tilde{\mathbf{V}}_t$ and the counter t form the sufficient statistic, i.e., $p(\theta|\mathcal{D}^{(t)}) \propto p(\theta|\tilde{\mathbf{V}}_t, t, \mathcal{D}^{(0)})$. They encapsulate the collected data and enrich the prior knowledge $\mathcal{D}^{(0)} = (\mathbf{V}_0, \Delta_0)$ to, see (1),

$$\begin{aligned} \mathbf{V}_t &= \sum_{\tau=0}^{t} \begin{pmatrix} y_\tau \\ z_\tau \end{pmatrix} \begin{pmatrix} y_\tau^\top z_\tau^\top \end{pmatrix} + \mathbf{V}_0 = \tilde{\mathbf{V}}_t + \mathbf{V}_0 \\ \Delta_t &= t + \Delta_0. \end{aligned} \tag{7}$$

The matrix $\mathbf{V}_0 > 0$ and degrees of freedom $\Delta_0 > \rho$ reflect the prior beliefs about θ quantified by conjugate (self-reproducing) Gauss-inverse-Wishart PDF, [1, 27]. The optimised choice of $\mathbf{V}_0$, Δ_0 is in Sec. 2.1. The matrix (7) is partitioned as

$$\mathbf{V}_t = \begin{pmatrix} \mathbf{V}_{y,t} & \mathbf{V}_{zy,t}^\top \\ \mathbf{V}_{zy,t} & \mathbf{V}_{z,t} \end{pmatrix}, \quad \mathbf{V}_{y,t} \in \mathbb{R}^{N,N}, \tag{8}$$

and the prior $\mathbf{V}_0$ is partitioned in the same way. The posterior likelihood with this prior is rewritten into

$$l(\mathcal{D}^{(t)},\theta)p(\theta) \quad (9)$$
$$= \frac{\exp\left[-\frac{1}{2}\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}(\mathbf{P}-\hat{\mathbf{P}}_t)^\top \mathbf{V}_{z,t}(\mathbf{P}-\hat{\mathbf{P}}_t)+\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda}_t\right)\right]}{(2\pi)^{N\frac{\Delta_t}{2}}|\boldsymbol{\Sigma}|^{\frac{\Delta_t}{2}}}.$$

The posterior parameter PDFs are

$$p(\mathbf{P}|\mathcal{D}^{(t)},\boldsymbol{\Sigma}^{-1}) = \frac{\exp\left[-\frac{1}{2}\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}(\mathbf{P}-\hat{\mathbf{P}}_t)^\top \mathbf{V}_{z,t}(\mathbf{P}-\hat{\mathbf{P}}_t)\right)\right]}{(2\pi)^{\frac{\rho N}{2}}|\mathbf{V}_{z,t}|^{-\frac{N}{2}}|\boldsymbol{\Sigma}|^{\frac{\rho}{2}}}$$
$$p(\boldsymbol{\Sigma}^{-1}|\mathcal{D}^{(t)}) = \frac{|\boldsymbol{\Lambda}_t|^{\frac{\Delta_t-\rho+N+1}{2}}|\boldsymbol{\Sigma}|^{-\frac{\Delta_t-\rho}{2}}\exp\left[-\frac{1}{2}\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda}_t\right)\right]}{(2^{\Delta_t-\rho+N+1}\pi^{\frac{N-1}{2}})^{\frac{N}{2}}\prod_{j=1}^{N}\Gamma\left(\frac{\Delta_t-\rho+N+2-j}{2}\right)}. \quad (10)$$

They exploit the analytical form of $J(\mathcal{D}^{(t)})$ (5), [1],

$$J(\mathcal{D}^{(t)}) = J(\mathbf{V}_t,\Delta_t) = (2\pi)^{\frac{\rho N}{2}}|\mathbf{V}_{z,t}|^{-\frac{N}{2}}|\boldsymbol{\Lambda}_t|^{-\frac{\Delta_t-\rho+N+1}{2}} \times (2^{\Delta_t-\rho+N+1}\pi^{\frac{N-1}{2}})^{\frac{N}{2}}\prod_{j=1}^{N}\Gamma\left(\frac{\Delta_t-\rho+N+2-j}{2}\right) \quad (11)$$

$$\hat{\mathbf{P}}_t = \mathbf{V}_{z,t}^{-1}\mathbf{V}_{zy,t},\ \boldsymbol{\Lambda}_t = \mathbf{V}_{y,t}-\mathbf{V}_{zy,t}^\top\mathbf{V}_{z,t}^{-1}\mathbf{V}_{zy,t}. \quad (12)$$

The maxima of these PDFs are achieved at $\mathbf{P}=\hat{\mathbf{P}}_t$ and $\hat{\boldsymbol{\Sigma}}_t^{-1}=\Delta_t\boldsymbol{\Lambda}_t^{-1}$, respectively. The PDFs (10) and the denominator (11) are exploited later on.

Often, regressors are (almost) collinear, [28]. Then, the sub-matrix $\mathbf{V}_{z,t}$ is poorly conditioned, which makes evaluations of the posterior PDF non-robust. This is resolved by updating the square root of the (positive semi-definite) matrix $\mathbf{V}_t$, [29]. The version suited to all learning parts is

$$\mathbf{V}_t = \mathbf{G}_t\mathbf{G}_t^\top = \mathbf{G}_{t-1}\mathbf{G}_{t-1}^\top + \begin{pmatrix}y_t\\ z_t\end{pmatrix}\left(y_t^\top z_t^\top\right), \quad (13)$$

where $\mathbf{G}_t$ is an upper triangular matrix. The square root is partitioned as $\mathbf{V}_t$ in (8), giving

$$\mathbf{V}_t = \mathbf{G}_t\mathbf{G}_t^\top = \begin{pmatrix}\mathbf{G}_{y,t}\mathbf{G}_{y,t}^\top+\mathbf{G}_{zy,t}\mathbf{G}_{zy,t}^\top & \mathbf{G}_{zy,t}\mathbf{G}_{z,t}^\top\\ \mathbf{G}_{z,t}\mathbf{G}_{zy,t}^\top & \mathbf{G}_{z,t}\mathbf{G}_{z,t}^\top\end{pmatrix}.$$

It provides the maximum a posteriori likelihood point estimates (12) of unknown $\mathbf{P}$, $\boldsymbol{\Sigma}^{-1}$. The maximum of (10) is reached by minimising over $\mathbf{P}$ the quadratic norm

$$\left\|\begin{pmatrix}\mathbf{G}_{y,t}^\top & \mathbf{O}\\ \mathbf{G}_{zy,t}^\top & \mathbf{G}_{z,t}^\top\end{pmatrix}\begin{pmatrix}-\mathbf{I}\\ \mathbf{P}\end{pmatrix}\right\|^2,$$

$\mathbf{O}$, $\mathbf{I}$ are zero and unit matrix, respectively. This, together with simple maximisation over $\boldsymbol{\Sigma}$, gives

$$\hat{\mathbf{P}}_t = \mathbf{G}_{z,t}^{-\top}\mathbf{G}_{zy,t}^\top,\quad \hat{\boldsymbol{\Sigma}}_t = \Delta_t\boldsymbol{\Lambda}_t \text{ with } \boldsymbol{\Lambda}_t = \mathbf{G}_{y,t}\mathbf{G}_{y,t}^\top. \quad (14)$$

The update $\mathbf{G}_{t-1} \to^{(y_t^\top\ z_t^\top)} \mathbf{G}_t$ equivalent to (13) involves one, numerically stable, orthogonal rotation, [30]. Moreover, it allows simple evaluations of determinants as products of diagonal entries of square root factors, $|\mathbf{V}_{z,t}| = \prod_{k=1}^{\rho} G_{kk,z,t}^2$ and $|\boldsymbol{\Lambda}_t| = \prod_{k=1}^{N} G_{kk,y,t}^2$. They are needed whenever the denominator $J(\cdot)$ (5) is used, see Secs. 2.2 and 2.3.

2.1. Optimal Hierarchical Choice of Prior PDF

For the inspected model (1), the conjugate prior is the Gaussian-inverse-Wishart PDF. The prior PDF is specified by $(\mathbf{V}_0,\Delta_0)$, where $\mathbf{V}_0>0$ is a prior extended information matrix and $\Delta_0>\rho$ is the prior degrees of freedom. Their influence on the estimation process is significant, especially on the structure estimation performed in off-line mode on $\bar{t}$ data records, see Sec. 2.2.

The paper [3] provides the optimal choice $(\mathbf{V}_0,\Delta_0)$ by maximising the marginal likelihood as a function of the explicitly expressed prior knowledge

$$l(\mathcal{D}^{(\bar{t})}|\mathbf{V}_0,\Delta_0) := \int_\Theta p(\mathcal{D}^{(\bar{t})}|\theta)p(\theta|\mathbf{V}_0,\Delta_0)d\theta = \frac{J(\mathbf{V}_{\bar{t}},\Delta_{\bar{t}})}{J(\mathbf{V}_0,\Delta_0)}$$

under the weakest common constraint that $(\mathbf{V}_0,\Delta_0)$ must make the prior PDF proper. The mentioned paper proves the existence of this extreme, gives its form, and verifies its positive influence.

Operationally, the construction form $\bar{t}$ data records is as follows. First, we map the data on the square root $\tilde{\mathbf{G}}_{\bar{t}}$ of the extended information matrix with zero prior

$$\tilde{\mathbf{V}}_{\bar{t}} = \sum_{\tau=1}^{\bar{t}}\begin{pmatrix}y_\tau\\ z_\tau\end{pmatrix}\left(y_\tau^\top z_\tau^\top\right)+\mathbf{O} = \tilde{\mathbf{G}}_{\bar{t}}\tilde{\mathbf{G}}_{\bar{t}}^\top.$$

It depends on the processed data $(y_\tau^\top z_\tau^\top)_{\tau=1}^{\bar{t}}$ only.

Then, we select a scalar tuning parameter $\mu\in(0,1)$. This scalar optional parameter controls the shrinkage strength of the resulting prior PDF. The optimal prior degrees of freedom Δ_0 are found by numerically solving the next nonlinear scalar equation with a unique solution

$$-\frac{1}{2}\ln(\mu) = \frac{\frac{1}{2}(1-\mu)\bar{t}}{\bar{t}+(1-\mu)\Delta_0}+h(\Delta_0)-h(\bar{t}+\Delta_0),$$

where $\bar{t}$ is the number of processed data, and $h(\Delta)$ is a function involving the digamma function $\Psi(\chi)=\frac{\mathrm{d}}{\mathrm{d}\chi}\ln(\Gamma(\chi)),\chi>0$, [31],

$$h(\Delta) := \ln(0.5\Delta)-\Psi(0.5\Delta).$$

With the value of Δ_0, the optimal prior matrix $\mathbf{V}_0$ is constructed analytically using the data-based statistics $\tilde{\mathbf{V}}_{\bar{t}}$ and μ as follows $\mathbf{V}_0 =$

$$\begin{pmatrix}\frac{\mu\Delta_0}{\bar{t}+(1-\mu)\Delta_0}\tilde{\mathbf{V}}_{y,\bar{t}}+\frac{\mu\bar{t}}{\bar{t}(1-\mu)+(1-\mu)^2\Delta_0}\tilde{\mathbf{V}}_{zy,\bar{t}}^\top\tilde{\mathbf{V}}_{z,\bar{t}}^{-1}\tilde{\mathbf{V}}_{zy,\bar{t}} & \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_{zy,\bar{t}}^\top\\ \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_{zy,\bar{t}} & \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_{z,\bar{t}}\end{pmatrix}$$

Experiments confirmed that the described choice of $(\mathbf{V}_0, \Delta_0)$ makes the structure and parameter estimation visibly more stable than the usual "textbook" options.

The prior derived in this way is intuitively interpreted as a shrinkage prior. It is equivalent to setting the prior point estimates (12) to

$$\begin{aligned}\hat{\mathbf{P}}_0 &= \tilde{\mathbf{V}}_{z,\bar{t}}^{-1}\tilde{\mathbf{V}}_{zy,\bar{t}}, \quad \mathbf{V}_{z,0} = \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_{z,\bar{t}} \\ \mathbf{\Lambda}_0 &= \frac{\mu\Delta_0}{\bar{t}+(1-\mu)\Delta_0}\left(\tilde{\mathbf{V}}_{y,\bar{t}} - \tilde{\mathbf{V}}_{zy,\bar{t}}^{\top}\tilde{\mathbf{V}}_{z,\bar{t}}^{-1}\tilde{\mathbf{V}}_{zy,\bar{t}}\right).\end{aligned}$$

This is a profound result. It effectively sets the prior mean of the coefficients as centred exactly on the data-based statistics. The prior beliefs about the uncertainty of the coefficients (given by the Kronecker product $\Delta_0^{-1}\mathbf{\Lambda}_0 \otimes \mathbf{V}_{z,0}^{-1}$, [1]) and the noise variance are scaled versions of the data-based estimates. A smaller μ implies a weaker prior that has a larger variance than the posterior, effectively down-weighting the prior's influence relative to the data.

The sub-matrices of the decomposed prior $\mathbf{G}_0\mathbf{G}_0^{\top}$, given that $\tilde{\mathbf{V}}_{\bar{t}} = \tilde{\mathbf{G}}_{\bar{t}}\tilde{\mathbf{G}}_{\bar{t}}^{\top}$ are

$$\begin{aligned}\mathbf{G}_{y,0} &= \sqrt{\frac{\mu\Delta_0}{\bar{t}+(1-\mu)\Delta_0}}\tilde{\mathbf{G}}_{y,\bar{t}}, \ \mathbf{G}_{zy,0} = \sqrt{\frac{\mu}{1-\mu}}\tilde{\mathbf{G}}_{zy,\bar{t}} \\ \mathbf{G}_{z,0} &= \sqrt{\frac{\mu}{1-\mu}}\tilde{\mathbf{G}}_{z,\bar{t}}.\end{aligned}$$

They are not recalculated because the posterior values matter.

2.2. Structure Estimation

The structure estimation is a crucial task in regression problems. It selects among ρ available regressors, stored in vectors $\bar{z}_t$ and possibly containing a_t, those significant for predicting y_t. Selecting appropriate regressors to z_t is computationally hard as there are $2^{\rho} - 1$ combinations of inclusion and exclusion of regressors from $\bar{z}_t$ into z_t.

The comparison of the competitors is often based on Akaike's information criterion and its improvements, [32]. Their approximate nature limits their performance. Therefore, an efficient, ideally non-approximate, Bayesian structure estimator should be used. The chapter by [2] offers such a maximum a posteriori probability structure estimator.

The maximised log-likelihood $\ln[l(u)] = \ln[J(\mathbf{V}_t(u), \Delta_t)] - \ln[J(\mathbf{V}_{\bar{t}}(u), \Delta_{\bar{t}})]$, see (11), acts on a vast, discrete set of "mutations" $\{u\}$ of indices of the inspected potential structures. The ρ-vector u contains units at those positions of $\bar{z}_t$, which appear in the currently considered z_t and zeros otherwise. For the ARX model, the log-likelihood on the structure mutations reads, see (6), (11)

$$\begin{aligned}&\ln[l(u)] = \\ &\quad\sum_{j=1}^{N}\ln\left[\Gamma(\tfrac{\Delta_t-\rho_u+N+1-j}{2})\right] - \ln\left[\Gamma(\tfrac{\Delta_{\bar{t}}-\rho_u+N+1-j}{2})\right] \\ &-\tfrac{N}{2}(\ln|\mathbf{V}_{z,t}(u)| - \ln|\mathbf{V}_{z,\bar{t}}(u)|) - \tfrac{\Delta_t-\rho_u+N}{2}N\ln|\mathbf{\Lambda}_t(u)| \\ &+\tfrac{\Delta_{\bar{t}}-\rho_u+N}{2}N\ln|\mathbf{\Lambda}_{\bar{t}}(u)| + constant, \qquad (15)\end{aligned}$$

where ρ_u is the number of units in u, $\mathbf{V}_{z,t}(u)$. $\mathbf{V}_{z,\bar{t}}(u)$ are the matrices gained from $\mathbf{V}_{\bar{z},t}$ and $\mathbf{V}_{\bar{z},\bar{t}}$ ($t > \bar{t}$, see Sec. 2.1) by omitting rows and columns corresponding to zeros in u, respectively. $\mathbf{\Lambda}_t(u) = \mathbf{V}_{y,t}(u) - \mathbf{V}_{zy,t}^{\top}(u)\mathbf{V}_{z,t}^{-1}(u)\mathbf{V}_{zy,t}(u)$, where $\mathbf{V}_{y,t}(u) = \mathbf{V}_{y,t}$ and in $\mathbf{V}_{zy,t}(u)$ the zero-related rows from $\mathbf{V}_{\bar{z}y,t}$ are omitted. $\mathbf{\Lambda}_{\bar{t}}(u)$ similarly arises from $\mathbf{V}_{\bar{t}}$.

As said, the square root version of the extended information matrix (13) makes the evaluation of determinants (15) computationally cheap for the triangular $\mathbf{G}$. The work [2] provides technical details on how to get the square roots of $\mathbf{V}_t(u), \mathbf{V}_{\bar{t}}(u)$ simply, from square roots of $\mathbf{V}_t, \mathbf{V}_{\bar{t}}$ collected for the widest inclusion of regressors from $\bar{z}_t$ into z_t.

For a realistic extent of potential regressors, the values of $\ln[l(u)]$ cannot be inspected completely, and they exhibit many local maxima. The work [2] performs local searches and finds the global maximum by their random restarts.

Here, we use a genetic algorithm [8] that operates on possible mutations and promotes exploration over exploitation by its setting.

Genetic algorithms, broadly, implement the optimisation inspired by natural evolution. They initiate a population of candidates from $\{u\}$, evaluate their fitness, here given by $\ln[l(u)]$ (15), and iteratively improve the candidates according to their fitness.

The adopted fitness selection procedure is called elitist selection, where the fittest candidate is simply the one that maximises the fitness function. Other methods, which emphasise exploration, would employ roulette wheel selection using softmax probabilities based on the fitness of the members in the given population.

The implemented version creates a new population by mutating the parent vector $\{u\}$ and creating a batch of candidates. This is done by randomly flipping each bit in $\{u\}$ with probability p_{mut}.

An offspring is created by crossing over two parents among the fittest population members, where the first k bits are contributed by the first parent, and the remaining $\rho - k$ bits are taken from the second parent. The

crossover point $k < \rho$ is randomly chosen with a uniform probability. This offspring is taken as a new parent $\{u\}$, from which a new population is created.

At the end of each iteration, the mutation probability p_{mut} is multiplied by a decay factor $\alpha_{decay} < 1$. This ensures convergence to local maxima.

Other variants are possible, e.g., a genetic algorithm may provide guesses initiating the algorithm in [2], and $\ln[l(u)]$ could include a non-uniform prior probability on $\{u\}$ if offered by an expert. For instance, the presence of a_t in z_t is quite improbable for small investors.

2.3. Data-Based Optimal Forgetting

The ARX model in Sec. 2 assumes constant $\mathbf{P}$, $\mathbf{\Sigma}$ and light-tails of the Gaussian distribution. These assumptions generally do not hold for real dynamic systems such as the financial markets.

Forgetting irrelevant knowledge provides a simple, flexible way of estimating varying parameter values. An advanced method of the adaptive choice of these forgetting factors based on the minimum relative entropy principle is taken from [16], where details on properties of the involved optimisation are available. It is applied to our case in a novel and useful way. By construction, it can discard an outlying data record and revert to the prior statistic; Sec. 4.3 verifies this on data with known parameters and shows that, with the prior used here, it does not track slow parameter drift.

Having a collection of possible relevant statistics $((\mathbf{V}_1, \Delta_1), (\mathbf{V}_2, \Delta_2), \dots)$ that determine the handled posterior Gauss-inverse-Wishart PDFs, the optimal forgetting weights $\varphi = (\varphi_1, \varphi_2, \dots)$, which enter the weighted geometric mean of the combined PDFs $(p(\theta|\mathbf{V}_j, \Delta_j))_{j\geq 1}$, are obtained by minimising the convex function of φ

$$F(\varphi) = D(\varphi||\varphi_0) - \ln\left[\int_\Theta \prod_{j\geq 1} p^{\varphi_j}(\theta|\mathbf{V}_j, \Delta_j) d\theta\right]. \quad (16)$$

$D(p||\bar{p}) = \int p(\chi) \ln[p(\chi)/\bar{p}(\chi)] \mathrm{d}\chi$ is the Kullback-Leibler divergence (KLD, [33]), φ_0 is the prior belief into the relevance of combined PDFs. It is a guess of the optimised belief φ. The parameter $\theta = (\mathbf{P}, \mathbf{\Sigma}^{-1})$ has the PDF, see (10),

$$p(\mathbf{P}, \mathbf{\Sigma}^{-1}|\mathbf{V}_j, \Delta_j) = p(\mathbf{P}|\mathbf{V}_j, \Delta_j, \mathbf{\Sigma}^{-1}) p(\mathbf{\Sigma}^{-1}|\mathbf{V}_j, \Delta_j).$$

The optimal belief, acting as forgetting, is the minimiser

$$\varphi^* \in \arg\min_{\varphi\in\Omega} F(\varphi), \quad \Omega = \left\{\varphi_j \geq 0, \sum_j \varphi_j = 1\right\}.$$

We combine the PDFs given by the sufficient statistics $(\mathbf{V}_j, \Delta_j)_{j=1}^{j=3}$ as those available at $t_1 = t, t_2 = t-1$, and $t_3 = \bar{t} < t$. The choice t_1, t_2 models the possible change of parameter values between time moments $t-1$ and t. The third term $t_3 = \bar{t}$ admits that the current data record (y_t, z_t) is such a big outlier that, for a stable estimation, we can only rely on the prior PDF given by the optimised $\mathbf{V}_0$, Δ_0.

The PDFs in the function (16) are weighted by factors φ_j. The PDF weighted by $\varphi_1 = \alpha$ is given to the updated statistic $(\mathbf{V}_t, \Delta)$, the PDF given by the old statistic $(\mathbf{V}_{t-1}, \Delta_{t-1})$ gets the weight $\varphi_2 = \beta$ and the regularisation PDF, given by $(\mathbf{V}_0, \Delta_0)$, is weighted by $\varphi_3 = \gamma = (1-\alpha-\beta)$. For his choice, the function (16) reads, cf. (7) and (11),

$$\begin{aligned} F(\alpha,\beta,\gamma) &= \alpha \ln[\alpha/\alpha_0] + \beta[\beta/\beta_0] + \gamma \ln[\gamma/\gamma_0] - \\ &\ln[J(\mathbf{V}_t, \Delta_t)] + \alpha \ln\left[J\left(\mathbf{V}_{t-1} + \begin{pmatrix} y_t \\ z_t \end{pmatrix}\left(y_t^\top z_t^\top\right), \Delta_{t-1} + 1\right)\right] \\ &+\beta \ln[J(V_{t-1}, \Delta_{t-1})] + \gamma \ln[J(V_0, \Delta_0)], \quad \text{with} \\ \mathbf{V}_t &= (\alpha+\beta)\mathbf{V}_{t-1} + \alpha \begin{pmatrix} y_t \\ z_t \end{pmatrix}\left(y_t^\top z_t^\top\right) + \gamma \mathbf{V}_0 \qquad (17) \\ \Delta_t &= (\alpha+\beta)\Delta_{t-1} + \alpha + \gamma\Delta_0. \end{aligned}$$

In the implementation, we can use the factorised matrices as $\mathbf{V}_t$ is a convex combination including the prior positive definite matrix, and thus is positive definite.

The definition of $(\mathbf{V}_t, \Delta_t)$ in (17) has a neat interpretation. The old statistic should hold higher, or at least the same, weight as the new data dyad. If neither the updated statistic nor the old one is sufficiently relevant, we revert to the "safe" prior $\mathbf{V}_0$. It also serves as a regularisation, preventing a complete loss of knowledge.

The update with simultaneous decomposition that gives (13) is made by orthogonal transformations using the RQ algorithm (equivalent to the QR algorithm, [29]). For the inspected (α, β, γ), it has the next form, which for the optimal values $(\alpha, \beta, \gamma)^*$ defines the new "forgotten" square root $\mathbf{G}_t$ of the extended information matrix

$$\mathbf{G}_t\mathbf{G}_t^\top = (\alpha+\beta)\mathbf{G}_{t-1}\mathbf{G}_{t-1}^\top + \alpha\begin{pmatrix} y_t \\ z_t \end{pmatrix}\left(y_t^\top z_t^\top\right) + \gamma\mathbf{G}_0\mathbf{G}_0^\top = \underbrace{\begin{pmatrix} \sqrt{\alpha+\beta}\mathbf{G}_{t-1} & \sqrt{\alpha}\begin{pmatrix} y_t \\ z_t \end{pmatrix} & \sqrt{\gamma}\mathbf{G}_0 \end{pmatrix}\mathbf{R}}_{\mathbf{G}_t}\underbrace{\mathbf{R}^\top\begin{pmatrix} \sqrt{\alpha+\beta}\mathbf{G}_{t-1}^\top \\ \sqrt{\alpha}\left(y_t^\top \quad z_t^\top\right) \\ \sqrt{\gamma}\mathbf{G}_0^\top \end{pmatrix}}_{\mathbf{G}_t^\top}.$$

$\mathbf{R}\mathbf{R}^\top = \mathbf{I}$, i.e., $\mathbf{R}$ is a unitary matrix, and $\gamma = 1-\alpha-\beta$. The optimal values give $\Delta_t = (\alpha^*+\beta^*)\Delta_{t-1} + \alpha^* + \gamma^*\Delta_0$.

3. Design of Optimal Policy

A relation of FPD, [34, 35], to LQR is exploited for the quantitative expression of decision aims, Sec. 3.1. Sec. 3.2 outlines a numerically safe, square root version of the LQR under the constraint $a_t \in \{\mathbf{1}^\top a = 1, a \geq 0\}$, with $\mathbf{1}$ being a vector consisting of units.

3.1. Preference Quantification for LQR via FPD

For simplicity of explanations, we model returns and regressors via the normal state-space form

$$s_t = \mathbf{A}_{t-1} s_{t-1} + \mathbf{B} a_t + \epsilon_t, \quad \epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_\epsilon), \tag{18}$$

with the non-minimal state s_t defined

$$s_t^\top = \begin{pmatrix} a_t^\top & a_{t-1}^\top & y_t^\top & x_t^\top & 1 \end{pmatrix}, \; s_t \in \mathbb{R}^{3N+\rho+1}.$$

The inclusion of consecutive actions helps to respect the transaction costs. The matrices in (18) are constructed using the certainty-equivalence approximation, which replaces the unknown parameter by its newest point estimate[2]. They read

$$\mathbf{A}_{t-1} = \begin{pmatrix} \mathbf{O} & \mathbf{O} & \dots & 0 \\ \mathbf{I} & \mathbf{O} & \dots & 0 \\ \hat{\mathbf{P}}_{t-1} & \mathbf{O} & 0 & \\ \mathbf{O} & \mathbf{O} & \hat{\mathbf{X}}_{t-1} & 0 \\ 0^\top & 0^\top & 0^\top & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \mathbf{I} \\ \mathbf{O} \\ \mathbf{O} \\ \mathbf{O} \\ 0^\top \end{pmatrix}, \tag{19}$$

$\mathbf{A}_{t-1} \in \mathbb{R}^{3N+\rho+1, 3N+\rho+1}$, $\mathbf{B} \in \mathbb{R}^{3N+\rho+1, N}$, and $\hat{\mathbf{X}}_{t-1}$ is the matrix that models the evolution of regressors $x_t = \hat{\mathbf{X}}_{t-1} x_{t-1} + \epsilon_{x,t}$, with $\epsilon_{x,t} \sim \mathcal{N}(0, \hat{\mathbf{\Sigma}}_x)$. Often, $\hat{\mathbf{X}}_{t-1}$ can be chosen as a constant matrix realising a shift in x_{t-1}, and new parts are modelled as a random walk. If need be, it can be estimated in the same way as the regression coefficients $\mathbf{P}$.

The covariance matrix $\mathbf{\Sigma}_\epsilon = \mathrm{diag}[\mathbf{O}, \mathbf{O}, \hat{\mathbf{\Sigma}}_{t-1}, \hat{\mathbf{\Sigma}}_x, 0]$.

As said, FPD generalises classical dynamic decision and stochastic control theories. It uses the same model and its learning as those theories, but instead of optimising expected performance index (reward, loss) it minimises KLD $D(p_r \| p^I)$ of the joint PDF $p_r(\mathcal{D}^H) = \prod_{t=1}^H m(s_t|a_t, s_{t-1}) r(a_t|s_{t-1})$ modelling the closed decision loop with the inspected decision rules r up to a horizon $H \in \mathbb{N}$ to its *ideal* counterpart $p^I(\mathcal{D}^H) = \prod_{t=1}^H m^I(s_t|a_t, s_{t-1}) r^I(a_t|s_{t-1})$ expressing the decision aims and constraints.

[2]This approximation is less harmful than it looks. We assume small investors whose individually selected actions have no influence on the markets. Thus, even exact dual policies [36] cannot enhance estimation quality. The no-influence setup also makes a priori the closed decision loop, bounded-input bounded-output, stable [37].

For the normal linear state-space model, normal linear decision rules (among which the optimal rules r^* lie) and analogically specified ideal, KLD becomes expectation of a sum of quadratic forms with the positions and kernels implied by the moments of $(m^I(s_t|a_t, s_{t-1}) r^I(a_t|s_{t-1}))_{t=1}^H$. All addends behave in the same way, so that we can focus on a single one.

We exploit the above simple observation for constructing the *ideal* PDF that aligns with the user's (investor's) goal to minimise the transaction costs

$$(a_t - a_{t-1})^\top \mathbf{D}_{t-1} (a_t - a_{t-1})$$

given by the known matrix $\mathbf{D}_{t-1} = \mathrm{diag}[d_{1,t-1}, \dots, d_{N,t-1}]$ where $d_{i,t-1} > 0$ is the given cost of the making transaction $a_{i,t-1} \to a_{i,t}$ with the asset $i \in \{1, \dots, N\}$. This is the goal unambiguously expressed by the ideal rule $r^I(a_t|s_{t-1}) \sim \mathcal{N}(a_{t-1}, \mathbf{D}_{t-1}^{-1})$.

At the same time, the user wants to maximise their return $a_t^\top y_t$. The corresponding ideal PDF is to modify the reality modelling PDF $m(s_t|a_t, s_{t-1})$. It should prefer configurations of s_t and a_t with a high return. Schwarz's inequality confirms that this wish is met by the ideal

$$m^I(s_t|a_t, s_{t-1}) \propto m(s_t|a_t, s_{t-1}) \exp[\omega a_t^\top y_t] =$$
$$\mathcal{N}_s(\mathbf{A}_{t-1} s_{t-1} + \mathbf{B} a_t, \mathbf{\Sigma}_\epsilon).$$

The optional scalar $\omega > 0$ balances two contradictory users' aims. Rules of the risk-averse policy are gained by considering the worst possible result

$$(\omega^*, r^*) \in \arg\max_{\omega > 0} \min_r D(p_r \| p^I). \tag{20}$$

Taking into account that PDF m and m^I entering (20) are two Gaussian distributions with identical covariance $\mathbf{\Sigma}_\epsilon$, differing in their expectation by $\omega \hat{\mathbf{\Sigma}}_{t-1} a_t$, we get

$$D(m \| m^I) = 0.5 \omega^2 a_t^\top \hat{\mathbf{\Sigma}}_{t-1} a_t.$$

Using this, the selected $r^I(a_t|s_{t-1})$, and the fact that the inner minimisation (20) can be found analytically, [34], the outer maximisation in (20) is equivalent to, cf. (14),

$$\begin{aligned} \omega^* &= \arg\max_{\omega > 0} \int \exp\Big[\\ &-0.5 \left((a_t - a_{t-1})^\top \mathbf{D}_{t-1} (a_t - a_{t-1}) + \omega^2 a_t^\top \hat{\mathbf{\Sigma}}_{t-1} a_t \right) \Big] \mathrm{d}a_t \\ &= \arg\max_{\omega > 0} \left(\ln |\mathbf{E}_{t-1}| + a_{t-1}^\top (\mathbf{D}_{t-1} - \mathbf{D}_{t-1} \mathbf{E}_{t-1}^{-1} \mathbf{D}_{t-1}) a_{t-1} \right) \\ &\mathbf{E}_{t-1} = \mathbf{D}_{t-1} + \omega^2 \hat{\mathbf{\Sigma}}_{t-1}, \;\; \hat{\mathbf{\Sigma}}_{t-1} = \Delta_{t-1} \mathbf{G}_{y,t-1} \mathbf{G}_{y,t-1}^\top. \end{aligned}$$

Differentiation of the minimised function with respect to ω provides the necessary (and in this case sufficient) condition for an extreme

$$\omega \mathrm{tr}\left(\mathbf{E}_{t-1}^{-1} \hat{\mathbf{\Sigma}}_{t-1} \right) + \omega a_{t-1}^\top \mathbf{D}_{t-1} \mathbf{E}_{t-1}^{-1} \hat{\mathbf{\Sigma}}_{t-1} \mathbf{E}_{t-1}^{-1} \mathbf{D}_{t-1} a_{t-1} = 0.$$

The optimal $\omega_{t-1} = \omega^*$ is evaluated using a simple root-finding algorithm. Both addends of this condition are non-negative for $\mathbf{D}_{t-1} \succ \mathbf{O}$, $\hat{\mathbf{\Sigma}}_{t-1} \succeq \mathbf{O}$ and $\omega > 0$, so it has no root in $(0, \infty)$ and the implementation returns an end point of the searched interval. Sec. 4.3 quantifies what this costs and treats the upper end $\omega_{\max}$ as the user-set quantity it effectively is. @@@MK: the outer maximisation (20) decreases monotonically in ω and needs revisiting.@@@ The partial quadratic loss reads[3]

$$\begin{pmatrix} a_t^\top & a_{t-1}^\top & x_{t-1}^\top \hat{P}_{t-1} \end{pmatrix} \mathbf{Q}_{t-1} \begin{pmatrix} a_t \\ a_{t-1} \\ \hat{P}_{t-1}^\top x_{t-1} \end{pmatrix}, \tag{21}$$

$$\mathbf{Q}_{t-1} = \begin{pmatrix} \mathbf{C}_{t-1} & -\mathbf{D}_{t-1} & -\frac{\omega_{t-1}}{2}\mathbf{I} \\ -\mathbf{D}_{t-1} & \mathbf{D}_{t-1}\mathbf{C}_{t-1}^{-1}\mathbf{D}_{t-1} & \frac{\omega_{t-1}}{2}\mathbf{D}_{t-1}\mathbf{C}_{t-1}^{-1} \\ -\frac{\omega_{t-1}}{2}\mathbf{I} & \frac{\omega_{t-1}}{2}\mathbf{C}_{t-1}^{-1}\mathbf{D}_{t-1} & \frac{\omega_{t-1}^2}{4}\mathbf{C}_{t-1}^{-1} \end{pmatrix},$$

where $\mathbf{C}_{t-1} = \mathbf{D}_{t-1} + 0.5\omega_{t-1}^2 \hat{\mathbf{\Sigma}}_{t-1}$ is positive definite, while $\mathbf{Q}_{t-1}$ is generally positive semi-definite. The externally supplied regressors are not penalised. Thus, $\mathbf{Q}_{t-1}$ (21) complemented appropriately by zeros determines the optimised expectation $E[s_t^\top \mathbf{Q}_{t-1} s_t | a_t, s_{t-1}]$.

3.2. Optimisation

Dynamic programming [24] is a universal tool for designing the optimal causal policy. It realises the backwards induction from the finite receding horizon $\tau + h$ to the current time τ. The minimising arguments in the next functional equation evolving the value function $v(s_t)$ (also known as loss-to-go) are optimal actions, $t = \tau + h, \dots, \tau$,

$$v(s_{t-1}) = \min_{a_t \in \Omega_{BE}} \mathrm{E}\left[\left\| \mathbf{Q}_{t-1}^{\frac{1}{2}} s_t \right\|^2 + v(s_t) \middle| a_t, \mathcal{D}^{(t-1)} \right].$$

In our case, it becomes a constrained sequential quadratic programming problem with

$$\begin{aligned} \Omega_{BE} &= \{a \in \mathbb{R}^N \mid \mathbf{1}^\top a = 1, a \geq 0\} \\ v(s_t) &= \|\mathbf{H}_t s_t\|^2 + \psi_t, \ \mathbf{H}_{\tau+h} = \mathbf{O}, \ \psi_{\tau+h} = 0, \end{aligned}$$

The matrix $\mathbf{H}_t$ is kept in an upper triangular form, and the scalar shift ψ_t is independent of the data. The adopted optimisation builds upon the semi-monotonic augmented Lagrangian algorithm for inequalities and equalities constrained minimisation, [7]. The procedure is divided into two phases, the outer and inner loop.

The outer loop of the algorithm minimises the Lagrangian in which the scalar λ cares about the considered equality constraint and propagates the estimated loss-to-go matrix $\mathbf{H}$ back in time.

$$\min_{a_t} \mathrm{E}\left[\left\| \begin{pmatrix} \mathbf{Q}_{t-1}^{\frac{1}{2}} & \mathbf{H}_t \end{pmatrix} s_t \right\|^2 \middle| a_t, \mathcal{D}^{(t-1)} \right] + \psi_t + \lambda(\mathbf{1}^\top a_t - 1).$$

The matrix $\begin{pmatrix} \mathbf{Q}_{t-1}^{\frac{1}{2}} & \mathbf{H}_t \end{pmatrix}$ is mapped by an orthogonal transformation to an upper triangular matrix, denoted $\mathbf{H}_t'$. Applying expectation gives

$$\begin{aligned} v(s_{t-1}) &= \min_{a_t} \left\| \mathbf{H}_t' \begin{pmatrix} \mathbf{B} & \mathbf{A}_{t-1} \end{pmatrix} \begin{pmatrix} a_t \\ s_{t-1} \end{pmatrix} \right\|^2 \\ &+ \mathrm{tr}(\mathbf{H}_t'^\top \mathbf{H}_t' \mathbf{\Sigma}^{\frac{\top}{2}} \mathbf{\Sigma}^{\frac{1}{2}}) + \psi_t + \lambda(\mathbf{1}^\top a_t - 1). \end{aligned}$$

The matrix $\mathbf{H}_t' \begin{pmatrix} \mathbf{B} & \mathbf{A}_{t-1} \end{pmatrix}$ is orthogonally rotated to the upper triangular form. The resulting matrix within the norm and the shift are

$$\tilde{\mathbf{H}}_t = \begin{pmatrix} \tilde{\mathbf{H}}_{t,a^*} & \tilde{\mathbf{H}}_{t,s} \\ \mathbf{O} & \tilde{\tilde{\mathbf{H}}}_{t-1} \end{pmatrix}, \ \psi_{t-1} = \psi_t + \mathrm{tr}(\mathbf{H}_t'^\top \mathbf{H}' \mathbf{\Sigma}^{\frac{\top}{2}} \mathbf{\Sigma}^{\frac{1}{2}}).$$

$\tilde{\mathbf{H}}_{t,a^*} \in \mathbb{R}^{N,N}$, $\tilde{\mathbf{H}}_{t,s} \in \mathbb{R}^{N,3N+\rho+1}$, $\tilde{\tilde{\mathbf{H}}}_{t-1} \in \mathbb{R}^{2N+\rho+1,3N+\rho+1}$.

The multiplier λ can be found explicitly. An additional rotation and (now unconstrained!) optimisation provide the updated matrix $\mathbf{H}$ with $e_N^\top = [0, \dots, 0, 1]$

$$\left\| \begin{pmatrix} \tilde{\tilde{\mathbf{H}}}_{t-1} \\ \mathbf{1} \frac{e_N^\top + \mathbf{1}^\top \tilde{\mathbf{H}}_{t,a^*}^{-1} \tilde{\mathbf{H}}_{t,s}}{\mathbf{1}^\top \tilde{\mathbf{H}}_{t,a^*} \mathbf{1}} \end{pmatrix} s_{t-1} \right\|^2 = \|\mathbf{H}_{t-1} s_{t-1}\|^2.$$

The second phase realises the inner loop of the algorithm, which concerns the inequality constraint. At the time τ, it uses an iterative algorithm that finds the optimal action that satisfies both constraints.

It is formally stated as in [7], see Sec. 6.2,

$$\min_{a \in \Omega_{BE}} f(a), \quad f(a) = 0.5 a^\top \tilde{\mathbf{H}}_a^\top \tilde{\mathbf{H}}_a a - (\tilde{\mathbf{H}}_s s)^\top \tilde{\mathbf{H}}_a a, \tag{22}$$

where $\tilde{\mathbf{H}}$ is the matrix from the last step of the Riccati recursion, gained from

$$\mathbf{H}_\tau' \begin{pmatrix} \mathbf{B} & \mathbf{A}_{\tau-1} \end{pmatrix}.$$

The augmented Lagrangian introduces a regularisation parameter ξ,

$$\mathcal{L}(a, \bar{\lambda}, \xi) = f(a) + \mathbf{W}^\top \bar{\lambda} + 0.5\xi \|\mathbf{W} a\|^2,$$

with gradient $\nabla \mathcal{L}(a, \bar{\lambda}, \xi)$

$$= (\tilde{\mathbf{H}}_a^\top \tilde{\mathbf{H}}_a + \xi \mathbf{W}^\top \mathbf{W}) a - (\tilde{\mathbf{H}}_s s)^\top \tilde{\mathbf{H}}_a a + \mathbf{W}^\top \bar{\lambda},$$

where $\bar{\lambda}$ is the multiplier λ from the last step of the Riccatti recursion.

The algorithm solves for the bound constraint in the inner loop and simultaneously updates the Lagrange

[3] Importantly, we assume the small investor with a negligible influence on the market, i.e. with $z_t = x_{t-1}$.

multiplier $\bar{\lambda}$ for the equality constraints in the outer loop. The multiplier $\bar{\lambda}$ and softmax[4] of the action vector a^*, computed previously, initiate the inner loop.

The Karush-Kuhn-Tucker conditions [38], which have to be satisfied to ensure the uniqueness of the feasible solution, are

$$g \geq \mathbf{O}, \quad g^\top (a - \mathbf{l}) = 0.$$

The augmented Lagrangian in the square root formalism converts this problem back to the original minimisation (22).

$$\mathcal{L}_{aug}(a, \bar{\lambda}, \xi) = f(a) = \frac{1}{2} a^\top \tilde{\mathbf{H}}_{aug}^\top \tilde{\mathbf{H}}_{aug} a - \mathbf{b}_{aug} a, \quad (23)$$

$$\text{where} \quad \tilde{\mathbf{H}}_{aug} = \begin{pmatrix} \tilde{\mathbf{H}}_a \\ \sqrt{\xi} \mathbf{1}^\top \end{pmatrix}, \quad \mathbf{b}_{aug} = -\tilde{\mathbf{H}}_a \tilde{\mathbf{H}}_s s - \mathbf{1}^\top \bar{\lambda} - \xi \mathbf{1}^\top,$$

with gradient $g = \tilde{\mathbf{H}}_{aug}^\top \tilde{\mathbf{H}}_{aug} a - \mathbf{b}_{aug}$. At each iteration in the inner loop, the action is updated via the gradient projection method

$$a^{(k+1)} = a^{(k)} - w^{(k)} g(a^{(k)}, \lambda^{(k)}, \xi^{(k)}), \; w^{(k)} = -\frac{g^\top h}{h^\top \tilde{\mathbf{H}}_a^\top \tilde{\mathbf{H}}_a h}$$

$$\text{where } h = \begin{cases} 0 & \text{if} \quad a \leq \mathbf{O} \wedge g \geq \mathbf{0} \\ -g & \text{otherwise} \end{cases}.$$

We update the Lagrange multiplier

$$\bar{\lambda}^{(k+1)} = \bar{\lambda}^{(k)} + \xi^{(k)} \mathbf{1}^\top a^{(k)}.$$

If the constraint violation $\mathbf{1}^\top a^{(k+1)} - 1$ is too large, we adjust the regularisation parameter $\xi^{(k+1)} = b\xi^{(k)}$. The book [7] recommends $\xi^{(0)} \in [0.5, 1.5]$, $b \in [2, 10]$ as the values for these hyperparameters. At last, we check for convergence.

4. Experimental Results

This section compares the proposed policy with standard benchmarks on paired portfolios, reports what each component contributes, and tests the adaptive forgetting. The protocol, the paired tests, the full ablation and sensitivity tables, the stress tests, three further market windows and the live test are in the complementary material, referred to below as Secs. S1–S11; the thesis [26] reports further variants. All numbers come from the released code (Sec. 5), which regenerates them from a single script.

[4] In Thompson's sampling, it may be the case that the matrix $\tilde{\mathbf{H}}_{t,a^*}$ in the Ricatti recursion is ill-conditioned for inversion, then we take the previous action as the initial guess.

4.1. Setup

From the S&P 500 constituents we took the 100 stocks with the highest average daily return in 2021 and, among them, the 20 least pair-wise correlated ones. We draw 200 portfolios of 8–12 stocks at random with a fixed seed and run every strategy, benchmark and ablation on exactly the same portfolios and days, so that all comparisons are paired. Each portfolio uses 750 trading days, 4 November 2022 – 31 October 2025: 125 days build the prior (Sec. 2.1), 125 days feed the structure estimation (Sec. 2.2), and the remaining 500 days are the test period. Every strategy starts from the uniform allocation and pays 0.2% of the value actually traded, the drift of the weights caused by the realised returns excluded. Differences across portfolios are tested with the Wilcoxon signed-rank test, a bootstrap 95% confidence interval of the mean difference and the win rate. The hyperparameters μ, α_0, β_0 were fixed by a 2^k design [39] on 2021–2022 data, disjoint from the test period. Secs. S1–S3 give the universe, the dates, the seeds, the cost model, the metrics and every benchmark in full, Sec. S4 the end-to-end algorithm with all user-set quantities and their origin.

4.2. Results on the 2023–2025 Test Period

Table 1 reports the means over the 200 portfolios; the paired tests and the median wealth paths are in Sec. S3.

With the corrected optimiser the Risk-Averse policy (RA) reaches an annualised Sharpe ratio of 0.34 and a total return of 1.11, against 0.66 and 1.27 for the daily rebalanced equal-weight portfolio (ΔSharpe -0.32, $p < 0.001$, RA better in 26% of the portfolios). The cause is in the last column: RA trades 15% of the portfolio per day and pays 15% of the initial wealth in costs over the two years, ten times the equal-weight portfolio, while before costs the two are comparable. On risk it does what its name promises: its maximum drawdown -25.5% is smaller than that of both equal-weight portfolios and far smaller than that of every mean-variance variant (-40% to -42%, $p < 0.001$), and only the 250-day minimum-variance portfolio is better.

The turnover has a single cause. The risk weight ω^* of Sec. 3.1 is returned as the upper end $\omega_{\max} = 10^3$ of the searched interval in all 10^5 decision steps, because the necessary condition for an extreme is a sum of terms that are positive for $\mathbf{D} \succ \mathbf{O}$, $\hat{\mathbf{\Sigma}} \succeq \mathbf{O}$, $\omega > 0$ and therefore has no root. At $\omega = 10^3$ the variance term of (21) exceeds the transaction-cost term by four orders of magnitude, so the cost effectively leaves the loss and the allocation is re-optimised from scratch every day. This is also the mechanism behind the small drawdowns.

The legacy softmax policy, which produced the tables of the previous version, keeps the weights within a daily turnover of 2.9% of uniform and is then indistinguishable from the rebalanced equal-weight portfolio in return and 0.03 below it in Sharpe ratio ($p < 0.001$). Our earlier claim of outperforming the uniform portfolio rested on charging that portfolio no costs and does not survive a paired test. The S&P 500 returned 1.62 at a Sharpe ratio of 1.58 over these days, which no strategy on this 20-stock universe approaches, equal-weight buy-and-hold included; we therefore make no claim against the index either.

strategy	return	Sharpe	max DD	turn./day
Risk-Averse (RA)	1.114	0.34	-25.5%	14.9%
RA, legacy softmax solver	1.257	0.63	-27.4%	2.9%
Equal-weight, rebalanced	1.273	0.66	-27.1%	1.6%
Equal-weight, buy & hold	1.357	0.77	-28.8%	0.0%
Inverse volatility (60d)	1.198	0.55	-26.4%	2.3%
Min-variance (120d)	1.128	0.41	-24.6%	2.8%
Min-variance (250d)	1.181	0.56	-23.0%	2.0%
Mean-variance (120d)	1.277	0.47	-41.6%	15.3%
Mean-variance (250d)	1.567	0.69	-39.8%	8.6%
Mean-var., cost-aware (120d)	1.362	0.56	-41.8%	1.1%
ARX-conditional mean-var.	0.732	-0.83	-40.6%	43.6%
S&P 500 (SPY) buy & hold	1.624	1.58	-18.8%	–

Table 1: Benchmark comparison on the 500 test days 2023-11-03 to 2025-10-31: total return, annualised Sharpe ratio, maximum drawdown and mean daily turnover, averaged over the 200 portfolios, all strategies on the same portfolios and days with a 0.2% proportional transaction cost. The annualised return and volatility, the Calmar ratio, the total cost paid and the medians over the portfolios are in Sec. S3 of the complementary material.

4.3. What the Components Contribute

Each component was removed or replaced in turn on 100 of the portfolios and the user-set quantities were varied around their values; Secs. S5–S7 give the tables.

Three components are indispensable, and for the same reason: dropping the FPD risk term (−72% of the capital), sampling θ instead of using the point estimate (−56%) and omitting the structure estimation (−49%) all widen or distort the predictive density that the loss then takes at face value. Three are inert: the corrected constrained solver and an independent SLSQP reference agree to $3 \cdot 10^{-7}$ in the allocation, the horizon $h = 5$ reproduces $h = 1$ to $7 \cdot 10^{-6}$, and replacing the data-based forgetting by a plain Bayesian update changes no metric measurably (Sharpe +0.04, $p = 0.13$), because the optimised weight of the prior stays between 0.5 and 0.8 on almost every day. Of the tuned quantities only the prior shrinkage μ matters, and 0.9 is not its best value: $\mu = 0.7$ keeps 15 of 137 regressors instead of 57 and gains 0.18 in Sharpe ratio ($p < 0.001$), still below the equal-weight portfolio. One ablation is a trap: with a flat prior the forecasts vanish and the constrained minimiser is exactly $1/N$, so the apparently best variant (Sharpe 0.72) is the equal-weight portfolio under another name.

Capping the interval of ω shows that no value of it rescues the design: at $\omega_{\max} = 1$ the policy reproduces the rebalanced equal-weight portfolio, and at $\omega_{\max} = 10$ or 100, as well as for three principled elicitations, the forecasts dominate the variance term and 46–73% of the capital is lost. The reason is the forecast itself, which over the test period has the same standard deviation as the realised returns (0.027), a correlation of −0.01 with them, a sign accuracy of 49.7% and twice the squared error of predicting zero. A loss that takes it at face value trades on noise, so the point forecast has to be shrunk, or the posterior uncertainty integrated into the design, before it enters the loss (Sec. S7).

On synthetic ARX data with known parameters, one-day crashes on all assets and an abrupt change of all coefficients (Sec. S8), the forgetting sets $\beta = 0$ and $\gamma \approx 1$ on the crash days: the record is discarded, the statistic reverts to the prior and the estimation error does not move, while a plain Bayesian update and exponential forgetting both absorb the outlier. This is the robustness claimed in Sec. 2.3; its price is that the mechanism protects by not learning, so the model does not re-learn after the parameter change either, which is why removing the forgetting costs nothing in the ablation. Repeating the comparison on the COVID crash, the 2022 bear market, a window posterior to all data used to design the method, and a 32-day live test on a broker feed (Secs. S9 and S10) leaves the ranking against the equal-weight portfolio and the smaller drawdown against mean-variance, which holds in 76–96% of the portfolios, unchanged in every window.

5. Conclusions

The paper gives a complete, internally consistent solution of the adaptive portfolio-optimisation problem, from the choice of the prior to a numerically safe constrained design, and evaluates it with a paired protocol that any alternative can be plugged into. The evaluation does not confirm that the solution pays back in its present form: after transaction costs the policy is dominated by the equal-weight portfolio, while it does deliver smaller drawdowns than mean-variance allocation and rejects outlying observations as designed. The ablation and sensitivity studies locate the two elements responsible, the elicitation of the risk weight ω, which has no interior optimum, and the strength of the optimal prior combined with the forgetting, which prevents the

model from learning beyond its initial estimate. Both are local and are the first items of future work; the released code makes the corresponding experiments a matter of changing one line.

An immediate enrichment concerns exploration, which may significantly help the large investors influencing the market: the randomised nature of the FPD optimal policy naturally adds a random component to the poorly exciting actions of the standard LQR solution. The effect can be enhanced by Thompson's sampling [40]: sampling the unknown parameter from the available PDF $p(\theta|\mathcal{D}^t)$, performing the LQR design with it and collecting the resulting loss-to-go $v^k(s)$ gives a (non)parametric estimate of its PDF, cf. [41]. Off-line, this predicts the achievable quality without closing the decision loop; on-line, it allows the design to be robustified and thus made even more risk-aware.

The most promising advancement is the use of more general models. Mixture ratios, [42], are our favourite as they offer an efficient way to address black swan events, [17]. It needs, however, to reach solution completeness. It requires elaborating an efficient structure estimation, safe numerical implementation for mixed (continuous and discrete) modelled random variables, and the corresponding, inevitably approximate, dynamic programming, [43].

Contribution, Competing Interest, Data and Software Access

T. Procházka contributed to writing, reviewing, software, data handling and experiments. M. Kárný provided conceptualisation, and contributed to writing and reviewing.

The authors have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Public data were used only. The complete code, the data preparation and the scripts that regenerate every table and figure of Sec. 4 are available at `https://github.com/ProchTomas/ResearchProject`.

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